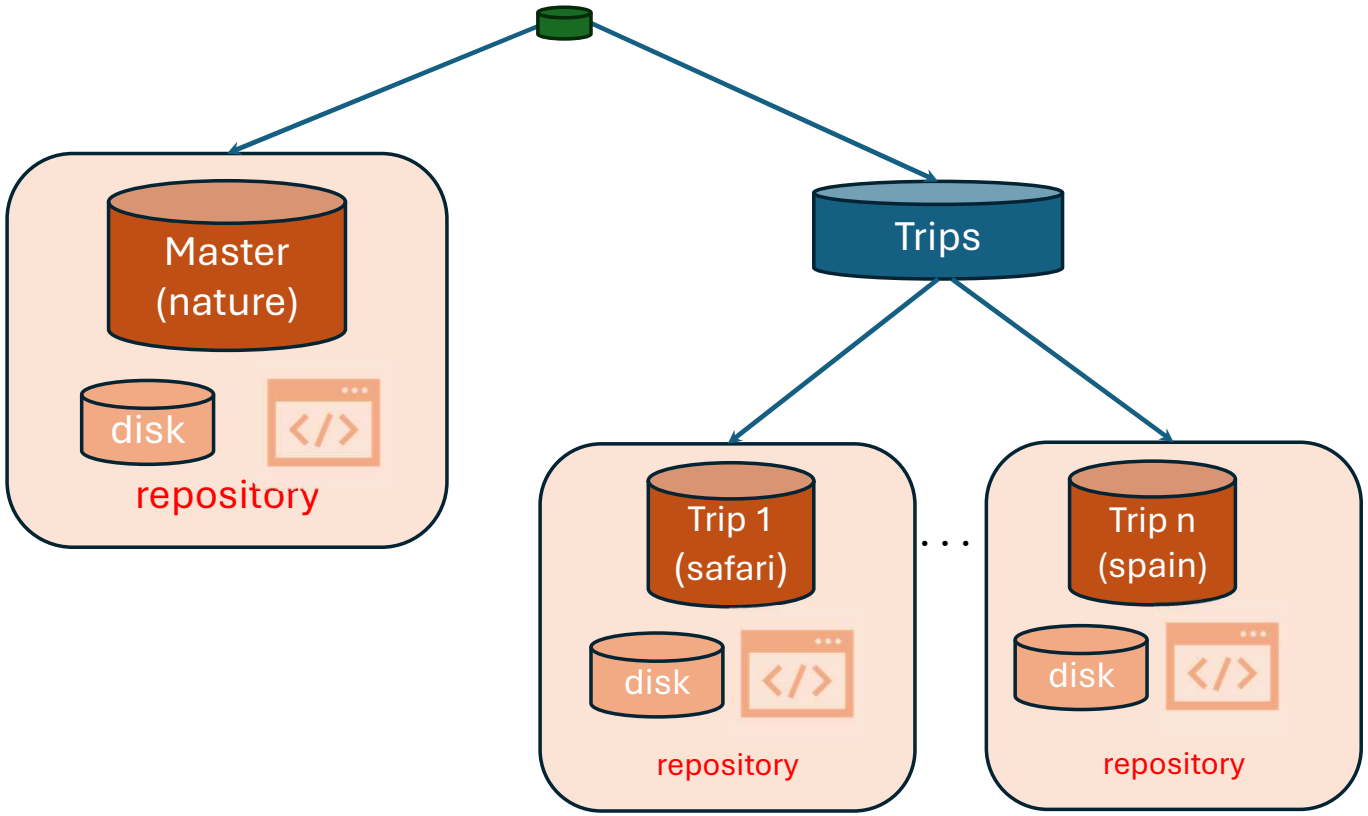
SpeciesViewer

Implementation notes

SpeciesViewer master and trip repository



Location on local servers

Master repository:
Disk: E:\dropbox\nature
URL: https:// .. /nature

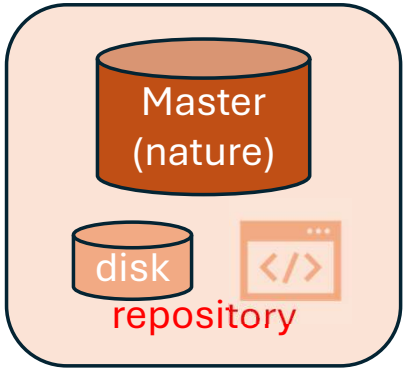
Trips have their own hierarchy:
Disk: E:\dropbox\trips\safaridb
URL: https:// .. /trips/safari

Disk: E:\dropbox\trips\spain2024
URL: https:// .. /trips/spain

The last qualifier is free-format, it is used as unique ID

Repository
(master or trip)

- **admin** has a blog administration for the repository.
- **biota** contains all species pictures, hierarchical organized. This a datastore, not aimed at publication
- **metadata** contains information all files and picture, as well as the species in our “cache” and what we actually have.
- **Web** contains a mirror of what is put on the website



admin/	• Blog_admin.csv
biota/	• animalia/ • plantae/ • fungi/
metadata/	• active/ • initialize/ • review/ (with subdirectories, see later)
Web/	• biota/ -- what is shown hierarchical on the website • biotablog/ – subdirecties every photo-blog • includes web/ – pre-installed helper html etc • summaries/ – statistics on the biota information

Files

The design is based on the concept that metadata is also be retrieved from files.

- RAWs from the camera
  - NEF or NRW from Nikon, HEIC from Apple or ORF or from Olympus
- XMP or DOP files with metadata (from Lightroom resp DxO PhotoLab)
- JPGs from the camera/iPhone, or from post-processing software

The database store in a trip is typically organized by day:

- The properties of all files are scanned with `exiftool` and saved in `data_files.csv`
- The concept of *picture* in SpeciesViewer is a set of files that are all based on a single camera click: `data_pictures.csv`. Typically this is at least a Raw and XMP file, optionally with JPGs, DOP and/or DNG (Adobe version of RAWs).
- `species.csv` contains the link between the species information an the pictures and files.
- `stats report yyyy-mm-dd.csv` summarizes the species found along with more statistics abot a directory is scanned.
- Statemachine.ini is a file to memorize which processing steps have been run on this project.

Configuration/descriptive files

SpeciesViewer adds some are .ini and .csv files in various places:

- `svconfig.ini` (variable name) – a file to describe the active repository.
- Blog entries require some text and a title for every project, which is in `blog.ini`. It is stored with the JPGs and RAWs (outside the repository)
- `admin/blog_admin.csv` – a property files that describes all blogs in a repository.
- `statemachine.ini` keeps track of progress of the process steps. It is stored in a file like `[name]/metadata/review/2024-06-26/statemachine.ini"`
- `trip.ini` – a property file that describes a trip repository, stored in the root of the trip (trips/[name]/trip.ini). Use it to give the repository a name, to provide a description, show a picture and a banner.
- `trips/Trip.csv` – like the blog_admin.csv all info from the trip.ini files.

Master and trip repositories (1)

The `svconfig.ini` defines the following directories/paths:

```
[repository]
BiotaDatebase=c:/laptop/trips/safaridb/
BiotaWebRoot=/trips/safari/
[project]
PictureDirectory=C:\CurrentPictures\2024\2024 Test Specviewer/2024-06-26
```

The same parameters work for the master and trips, but do not allow for more customization for trips, or for trips to reference existing species. This done with

```
BiotaMasterDatabase=c:/laptop/nature/
```

Based on this link, the master is available as well as (implied) the trips directory (all but last qualifier `trips/` from the `BiotaWebRoot`), and the current trip name (`safari/`)

Master and trip repositories (2)

Most of the logic is in the MyProject initialization, that reads the active svconfig.ini. `BiotaDatebase` and `BiotaWebRoot` are given.

When a `BiotaMasterDatabase` is specified, the following can be obtained:

```
ReferenceSpecies = join(BiotaMasterDatabase,
                        metadata\active\species.csv)
     .. Or explicit in svconfig.ini
RepositoryBanner = trip.ini:trip_banner
      .. Or “hardcoded” in installation section
RepositoryPicture = trip.ini:trip_picture
                    (path relative to web/)
```

sv1MakeRepository

Create a new repository with standard directory and FTP structures

Pre-reqs:

- svconfig.ini required
- Directory best made with trip.ini file in the root

Utilities

Utility	Works on ini	Needs edits?*	Purpose
svr1AddSpecies	Both	No	Helper. Add species to speciescollection.csv based on list in admin/speclist.csv
svr2CountSpecies	Trip	No	Stats. Count the number of species that are not yet in master
svr3StatusProjects	Trip	No	Stats. Report project steps that are run for trip
svr4UpdateTrips	Trip	trip.ini	Needed for each trip. Update the metadata for trip and update website.
svr5MergeSpeciesCollections		Yes	Needed for new English/Dutch names. Merge the species from 2 lists, eg master and trip
svr6MergeBlogsForMaster	Master	No	Needs at least once for each trip, blog count update not automatic. Create a different blog report for master, blogs per trips rather than all blogs
svr7DeleteFiles	Both		Clean duplicate files. (does not yet exist) Delete files that have been copied to master

* /trips/ is fixed and assumed to the high level directory for trips, that are second level directories